

PhD Analysis Examination

August 2024

Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Attempt SIX problems.

1. For any two of the following, either give an example or explain why no example exists.
 - a) a sequence of Lebesgue measurable functions on $\mathbb{R}$ which converges in measure but not Lebesgue a.e.
 - b) a closed set $E \subset [0, 1]$ with positive Lebesgue measure, but which contains no open intervals
 - c) a decreasing sequence of nonnegative measurable functions on $\mathbb{R}$ such that $\lim \int f_n \neq \int \lim f_n$
2. State Tonelli's theorem, and give an example to show that the σ -finiteness hypothesis is necessary.
3. a) State the Lebesgue-Radon-Nikodym theorem. b) State and prove a version of the chain rule for Radon-Nikodym derivatives.
4. Suppose that $f : \mathbb{R} \rightarrow \mathbb{C}$ is a Lebesgue measurable function with the property that $fg \in L^1(\mathbb{R})$ for every $g \in L^2(\mathbb{R})$. Must f belong to $L^2(\mathbb{R})$? Prove, or give a counterexample.
5. a) Show that for every integer $n \geq 1$, there exists a function $f_n \in L^2[0, 1]$ such that for every polynomial p of degree at most n ,

$$p(0) = \int_0^1 p(x) f_n(x) dx.$$

- b) Prove that there is *no* L^2 function f such that the above holds for *all* polynomials p (independent of the degree n).
6. State the Baire category theorem. Prove that if $\mathcal{X}$ is an infinite-dimensional Banach space, then $\mathcal{X}$ does not have a countable basis.
 7. Let $\mathcal{X}$ be a Banach space and $\mathcal{Y}$ a normed vector space. Let (T_n) be a sequence of bounded linear operators from $\mathcal{X}$ to $\mathcal{Y}$. Suppose that $\lim_n T_n x$ exists for each $x \in \mathcal{X}$. Prove that the linear transformation $Tx := \lim_n T_n x$ is bounded from $\mathcal{X}$ to $\mathcal{Y}$.
 8. For each of the following statements, determine if it is true or false, and sketch a proof of your claim.

- a) If ν is a signed measure and E, F are measurable sets with $\nu(E) \geq 0$ and $\nu(F) \geq 0$, then $\nu(E \cup F) \geq 0$.
- b) If $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces and $E \subset X, F \subset Y$ are sets such that $E \times F$ is $\mathcal{M} \otimes \mathcal{N}$ -measurable, then $E \in \mathcal{M}$ and $F \in \mathcal{N}$.
- c) $L^1(\mathbb{R})$ is reflexive.